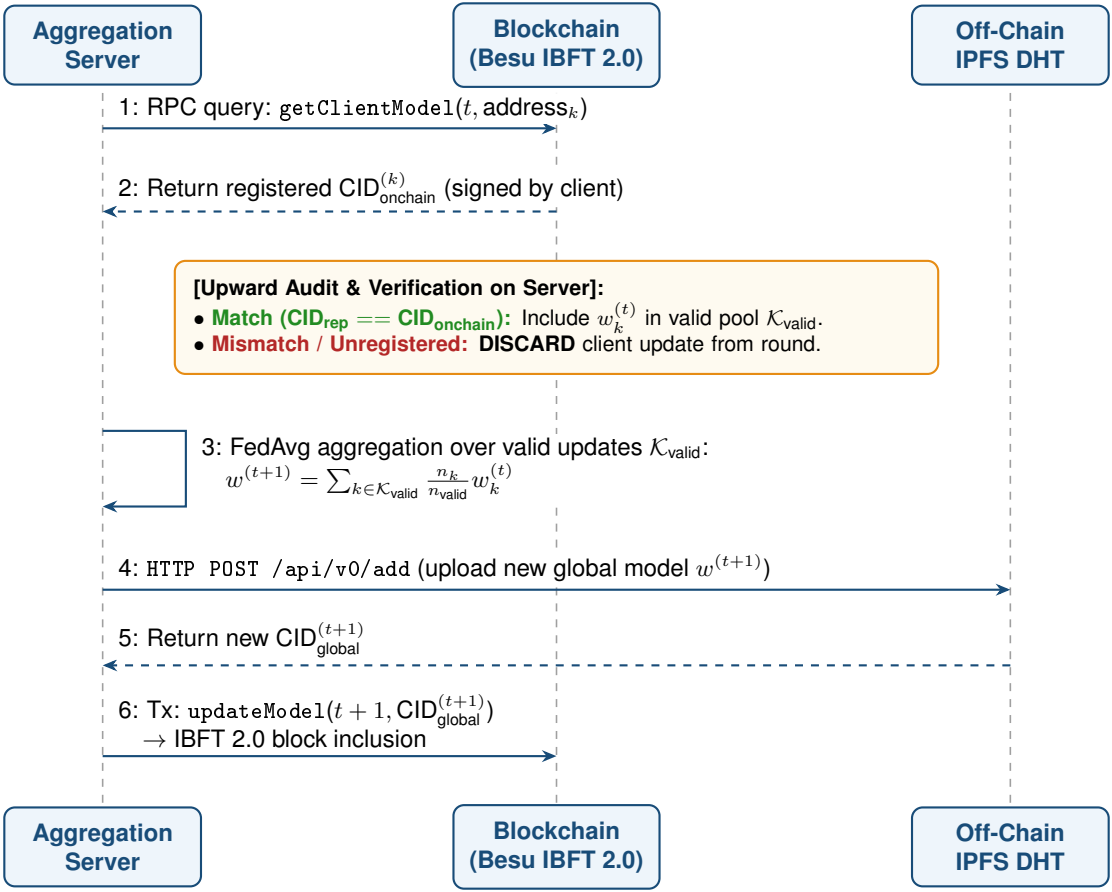

Aggregation Server
Blockchain (Besu IBFT 2.0)
Off-Chain IPFS DHT
1: RPC query: getClientModel(t, address$_k$)
2: Return registered CID$^{(k)}_{\text{onchain}}$ (signed by client)
[Upward Audit & Verification on Server]:
• Match (CID$_{\text{rep}}$ == CID$_{\text{onchain}}$): Include $w_k^{(t)}$ in valid pool $\mathcal{K}_{\text{valid}}$.
• Mismatch / Unregistered: DISCARD client update from round.
3: FedAvg aggregation over valid updates $\mathcal{K}_{\text{valid}}$:
$w^{(t+1)} = \sum_{k \in \mathcal{K}_{\text{valid}}} \frac{n_k}{n_{\text{valid}}} w_k^{(t)}$
4: HTTP POST /api/v0/add (upload new global model $w^{(t+1)}$)
5: Return new CID$^{(t+1)}_{\text{global}}$
6: Tx: updateModel($t+1$, CID$^{(t+1)}_{\text{global}}$)
→ IBFT 2.0 block inclusion
Aggregation Server
Blockchain (Besu IBFT 2.0)
Off-Chain IPFS DHT